

D900

Methods of Teaching Secondary Mathematics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A first-year teacher who earned a math degree with honors can solve every Algebra II problem, but students say her explanations are hard to follow. She also coaches the chess club. What should she develop most?

- A. Classroom management routines.
- B. Pedagogical content knowledge: common misconceptions, representations, and sequencing.
- C. Nothing; strong content knowledge is always enough to teach well.
- D. More advanced mathematics coursework.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A student in a class of 29 is stuck finding the rate of change from a table. The teacher, who has taught for eight years, wants to ask a focusing question. Which question fits?

- A. "You subtract the y-values first, right?"
- B. "Let me show you how to do the first one."
- C. "What do you notice about how y changes each time x goes up by 1?"
- D. "Is the rate 2 or 3?"

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

An observer notes that a teacher asked a chain of short questions, each with one right answer, until the student reached the teacher's solution. What questioning pattern is this?

- A. Socratic questioning, which always builds understanding.
- B. Wait time.
- C. Focusing.
- D. Funneling.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A class of 32 with widely mixed readiness has one goal: solve systems of linear equations. The lesson is scheduled right after an assembly. Which plan best differentiates?

- A. Group students by ability for the whole year.
- B. Give struggling students arithmetic worksheets instead.
- C. Tiered tasks on systems, with graphs and guided steps for some and open contexts for others.
- D. Give ready students 20 extra problems of the same type.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

After strong core instruction, a universal screening given on laptops in the media center shows four students below benchmark on fraction operations. The screening took 20 minutes. What is the best next step?

- A. Provide Tier 2 small-group intervention and monitor progress.
- B. Refer all four students for special education evaluation now.
- C. Reteach the whole class the same way.
- D. Move them straight to Tier 3 individualized intervention.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A student who did not respond to small-group support now receives daily, one-on-one intensive intervention. Which MTSS tier is this?

- A. Tier 2.
- B. Special education, because Tier 3 and special education are the same thing.
- C. Tier 1.
- D. Tier 3.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A student's IEP was updated at a meeting in March. For a geometry test, the teacher considers several changes. The school uses a block schedule. Which change is an accommodation rather than a modification?

- A. Grading against an easier standard.
- B. Extended time on the same test.
- C. Any change listed in an IEP is a modification.
- D. Removing all proof items.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which change is a modification rather than an accommodation?

- A. The student solves only one-step equations while the class solves multi-step equations.
- B. Directions are read aloud.
- C. A calculator for arithmetic on an algebra task.
- D. Preferential seating near the board.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A math teacher and a special education teacher share a class of 30 in a room with two whiteboards. They want a lower student-teacher ratio, so each teaches the same lesson to half the class at the same time. Which model is this?

- A. Station teaching.
- B. One teach, one assist.
- C. Team teaching, since co-teaching always means both lead the whole class.
- D. Parallel teaching.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

One co-teacher pulls a small group to pre-teach vocabulary while the other teaches the rest of the class. Which model is this?

- A. Station teaching.
- B. Alternative teaching.
- C. Parallel teaching.
- D. One teach, one observe.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A teacher who recently moved from another state wants to humanize a unit on linear functions. The unit runs for three weeks in the fall. Which plan fits?

- A.** Lower expectations so the unit feels less stressful.
- B.** Teach one standard method only to avoid confusion.
- C.** Have students model rates from their own communities and share different strategies, including in home languages.
- D.** Put posters of famous mathematicians on the wall.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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12.

Students are three minutes into a new area problem. A student wearing a soccer jacket says, "I'm stuck." The room is warm because the heat is broken. What should the teacher do?

- A.** Ask, "What have you tried, and what does the problem ask you to find?"
- B.** Show the first two steps of the solution.
- C.** Tell the student to wait until the teacher explains at the end.
- D.** Switch to an easier task, since struggle is always harmful.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A first-year teacher who earned a math degree ~~with honors~~ can solve every Algebra II problem, but students say her explanations are hard to follow. ~~She also coaches the chess club.~~ What should she develop most?

Target word: which kind of knowledge is missing **Decides:** She knows the math but not how students learn it.

A	Confusion set	The issue is explanations, not behavior.
B	Correct	PCK connects her math knowledge to how students make sense of it.
C	Absolute language	Knowing math and teaching it are different skills.
D	Adjacent category	Her content knowledge is already strong.

2. Answer C Concept 2

A student ~~in a class of 29~~ is stuck finding the rate of change from a table. The teacher, ~~who has taught for eight years,~~ wants to ask a focusing question. Which question fits?

Target word: focusing question **Decides:** A focusing question invites the student's own reasoning.

A	Inverted operation	Leads the student down the teacher's path instead of drawing out theirs.
B	Adjacent category	A demonstration, not a question.
C	Correct	Open, and builds on what the student can see.
D	Confusion set	A funneling choice with one expected answer.

3. Answer D Concept 2 · transfer

An observer notes that a teacher asked a chain of short questions, each with one right answer, until the student reached the teacher's solution. What questioning pattern is this?

Target word: name the pattern **Decides:** Step-by-step leading to the teacher's path.

A	Absolute language	A chain of leading questions is funneling, and no pattern always works.
B	Adjacent category	Wait time is a pause, not a question pattern.
C	Inverted operation	Focusing keeps the student's thinking at the center.
D	Correct	The teacher did the thinking; the student filled in blanks.

4. Answer C Concept 3

A class of 32 with widely mixed readiness has one goal: solve systems of linear equations. ~~The lesson is scheduled right after an assembly.~~ Which plan best differentiates?

Target word: sound differentiation **Decides:** Same goal, different supports.

A	Absolute language	Fixed tracking is not responsive, lesson-level differentiation.
B	Dropped element	Drops the shared learning goal.
C	Correct	Every student works toward the same goal with fitting support.
D	Adjacent category	More of the same is not deeper work.

5. Answer A Concept 4

After strong core instruction, a universal screening given on laptops in the media center shows four students below benchmark on fraction operations. ~~The screening took 20 minutes.~~ What is the best next step?

Target word: next MTSS step **Decides:** A few students need more than Tier 1.

A	Correct	Targeted support with data comes next for a few students.
B	Adjacent category	Skips the tiered interventions that come first.
C	Dropped element	Ignores that only four students need targeted help.
D	Confusion set	Tier 3 is for students who do not respond to Tier 2.

6. Answer D Concept 4 • transfer

A student who did not respond to small-group support now receives daily, one-on-one intensive intervention. Which MTSS tier is this?

Target word: name the tier **Decides:** Intensive and individualized.

A	Confusion set	Tier 2 is small-group support, which the student already had.
B	Adjacent category	Tier 3 is an intervention level, not a special education placement.
C	Inverted operation	Tier 1 is core instruction for everyone.
D	Correct	Tier 3 is the most intensive, individualized level.

7. Answer B Concept 5

A student's IEP ~~was updated at a meeting in March.~~ For a geometry test, the teacher considers several changes. ~~The school uses a block schedule.~~ Which change is an accommodation rather than a modification?

Target word: accommodation **Decides:** The content and standard stay the same.

A	Adjacent category	Lowering the standard is a modification.
B	Correct	Changes how the student shows learning, not what is learned.
C	Absolute language	IEPs list both kinds.
D	Confusion set	Removing content changes what is expected: a modification.

8. Answer A Concept 5 • transfer

Which change is a modification rather than an accommodation?

Target word: identify a modification **Decides:** It lowers the learning expectation.

A	Correct	The content expectation itself is reduced.
B	Confusion set	Changes access, not content: an accommodation.
C	Inverted operation	Supports access when arithmetic is not the skill tested: an accommodation.
D	Adjacent category	An environmental accommodation.

9. Answer D Concept 6

A math teacher and a special education teacher share a class of 30 ~~in a room with two whiteboards~~. They want a lower student-teacher ratio, so each teaches the same lesson to half the class at the same time. Which model is this?

Target word: co-teaching model **Decides:** Same content, two halves, at once.

A	Confusion set	In stations, each teacher runs different content and groups rotate.
B	Adjacent category	Only one teacher leads; the other circulates.
C	Absolute language	Team teaching is only one of several models.
D	Correct	Each teacher delivers the same content to half the class.

10. Answer B Concept 6 · transfer

One co-teacher pulls a small group to pre-teach vocabulary while the other teaches the rest of the class. Which model is this?

Target word: name the model **Decides:** Small group, specific purpose.

A	Adjacent category	Stations rotate all students through each teacher.
B	Correct	One teacher takes a small group for a targeted purpose.
C	Confusion set	Parallel splits the class evenly for the same lesson.
D	Dropped element	Here the second teacher is teaching, not observing.

11. Answer C Concept 7

A teacher ~~who recently moved from another state~~ wants to humanize a unit on linear functions. ~~The unit runs for three weeks in the fall.~~ Which plan fits?

Target word: a humanizing practice **Decides:** Connect to students' lives and value their ways of thinking with high expectations.

A	Confusion set	Humanizing keeps expectations high.
B	Inverted operation	Narrows whose thinking counts.
C	Correct	Honors identities and multiple ways of reasoning while keeping rigor.
D	Adjacent category	Decoration alone does not change how students experience math.

12. Answer A Concept 8

Students are three minutes into a new area problem. A student ~~wearing a soccer jacket~~ says, "I'm stuck." ~~The room is warm because the heat is broken.~~ What should the teacher do?

Target word: support productive struggle **Decides:** Keep the thinking with the student.

A	Correct	Builds on what the student knows without taking over.
B	Adjacent category	Rescuing removes the struggle that builds understanding.
C	Dropped element	Leaves the student with no way forward.
D	Absolute language	Productive struggle supports learning.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	12	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Content knowledge vs. pedagogical content knowledge	Knowing the mathematics is different from knowing how students learn it: common misconceptions, useful representations, and how to sequence ideas.	Assuming strong content knowledge is enough to teach well.	A teacher who can do the math but whose students cannot follow.
2	Focusing vs. funneling questions	Focusing questions press students to explain and extend their own thinking; funneling questions lead them step by step to the teacher's path.	Counting a chain of short leading questions as good questioning.	A series of teacher questions that each have one expected answer.
3	Differentiation keeps the goal	Differentiate process, product, or support while holding every student to the same grade-level learning goal.	Giving strong students more of the same, or giving struggling students a lower goal.	A class with mixed readiness and one lesson objective.
4	MTSS tiers	Tier 1 is core instruction for all; Tier 2 is targeted small-group intervention with progress monitoring; Tier 3 is intensive, individualized intervention.	Jumping straight to a special education referral or to Tier 3.	Screening data showing a few students below benchmark.
5	Accommodation vs. modification	An accommodation changes how a student accesses or shows learning; a modification changes what the student is expected to learn.	Calling any IEP change an accommodation.	A list of supports and "which is an accommodation?"
6	Co-teaching models	One teach-one assist, station, parallel (each teaches the same content to half the class), alternative (one takes a small group for a specific purpose), and team teaching.	Mixing up parallel and station teaching, or assuming co-teaching always means both lead the whole class.	A description of how two teachers split the room.
7	Humanizing mathematics	Humanizing approaches connect mathematics to students' identities and communities, value multiple strategies and languages, and keep expectations high.	Treating humanizing as decoration or as lowering rigor.	"Make the unit more relevant / humanizing."
8	Supporting productive struggle	When students are stuck, ask questions that build on what they know instead of taking over the thinking.	Rescuing students by showing the steps.	"A student says, 'I'm stuck.'"

Course scope (public description): Methods of Teaching Secondary Mathematics is an in-depth exploration of secondary mathematics content and pedagogy, distinguishing between subject mastery and effective teaching practices. This course will prepare students to teach and reflect on secondary mathematics lessons that emphasize the use of effective teaching practices, including differentiation, questioning techniques, and curriculum development. The course also covers strategies for presenting mathematical material at the appropriate level, utilizing humanizing approaches, co-teaching, Multi-Tiered Systems of Support (MTSS), accommodations, and inclusion. This course does not have any required prerequisites.